

Continuum Mechanics

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Chapter 04

Kinematics
Part 4



Change of Variables

Result 4.8 (Change of Variables)

Let

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

be a motion. For every $t \in [0, \infty)$, set

$$B_t = \boldsymbol{\phi}(B, t) = \boldsymbol{\phi}_t(B).$$

Suppose that q is an arbitrary spatial scalar field on B_t, and let $\Omega \subseteq B$ with

$$\Omega_t = \boldsymbol{\phi}(\Omega, t) = \boldsymbol{\phi}_t(\Omega) \subseteq B_t.$$

Then

$$\int_{\Omega_t} q(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega} q\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) \, \det\big(F(\mathbf{X}, t)\big) \, \mathrm{d}V_{\mathrm{X}},$$

where

$$F(\mathbf{X}, t) = \nabla^{\mathrm{X}} \boldsymbol{\phi}(\mathbf{X}, t).$$

Result 4.8 (Cont.)

Recall that the material description of q is

$$q_m(\mathbf{X}, t) = q\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big).$$

Thus the change of variables formula is

$$\int_{\Omega_t} q(\mathbf{x}, t)\ \mathrm{d}V_{\mathrm{x}} = \int_{\Omega} q_m(\mathbf{X}, t)\, \det\big(F(\mathbf{X}, t)\big)\ \mathrm{d}V_{\mathrm{X}}.$$

Proof.

See any good book on advanced calculus. □

Definition (Jacobian Field)

The scalar field

$$J(\mathbf{X}, t) := \det\big(F(\mathbf{X}, t)\big)$$

is called the *Jacobian field* for the deformation $\boldsymbol{\phi}_t$.

Corollary (Local Volume Preservation at One Time)

Suppose that, for a motion

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3,$$

there is a time $t_0 \in (0, \infty)$ such that

$$J(\mathbf{X}, t_0) = 1, \qquad \mathbf{X} \in \Omega.$$

Then

$$\text{Vol}(\Omega) = \text{Vol}(\Omega_{t_0}).$$



Proof.

By Result 4.8,

$$\begin{aligned}
\text{Vol}(\Omega_{t_0}) &= \int_{\Omega_{t_0}} \mathrm{d}V_{\mathrm{x}} \\
&= \int_{\Omega} J(\mathbf{X}, t_0)\ \mathrm{d}V_{\mathrm{X}} \\
&= \int_{\Omega} \mathrm{d}V_{\mathrm{X}} \\
&= \text{Vol}(\Omega).
\end{aligned}$$





Result 4.9 (Evolution of the Jacobian)

Suppose

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

is a motion with spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ and deformation gradient $F(\mathbf{X}, t)$. Then

$$\frac{\partial}{\partial t} \det\big(F(\mathbf{X}, t)\big) = \det\big(F(\mathbf{X}, t)\big) \big[\nabla^{\mathrm{x}} \cdot \mathbf{v}\big]_m(\mathbf{X}, t).$$

Equivalently,

$$\frac{\partial J}{\partial t} = J \big[\nabla^{\mathrm{x}} \cdot \mathbf{v}\big]_m.$$



Proof.

Result 2.23 gives

$$\begin{aligned}\frac{\partial}{\partial t}\det\big(F(\mathbf{X},t)\big) &= \det\big(F(\mathbf{X},t)\big)F(\mathbf{X},t)^{-T} : \frac{\partial F}{\partial t}(\mathbf{X},t) \\ &= \det\big(F(\mathbf{X},t)\big)\,\mathrm{tr}\left(F(\mathbf{X},t)^{-1}\frac{\partial F}{\partial t}(\mathbf{X},t)\right) \\ &= \det\big(F(\mathbf{X},t)\big)\,\mathrm{tr}\left(\frac{\partial F}{\partial t}(\mathbf{X},t)F(\mathbf{X},t)^{-1}\right). \qquad (4.22)\end{aligned}$$

The final equality uses the cyclic property of the trace.

Proof (Cont.)

Recall that the material velocity satisfies

$$\mathbf{V}(\mathbf{X}, t) = \mathbf{v}_m(\mathbf{X}, t) = \frac{\partial}{\partial t}\boldsymbol{\phi}(\mathbf{X}, t).$$

In component form,

$$V_i(\mathbf{X}, t) = [\mathbf{V}(\mathbf{X}, t)]_i = [\mathbf{v}_m(\mathbf{X}, t)]_i = \frac{\partial \phi_i}{\partial t}(\mathbf{X}, t).$$

Also,

$$F_{ij}(\mathbf{X}, t) = \frac{\partial \phi_i}{\partial X_j}(\mathbf{X}, t).$$

Therefore, equality of the mixed partial derivatives gives

$$\frac{\partial F_{ij}}{\partial t}(\mathbf{X}, t) = \frac{\partial}{\partial X_j} V_i(\mathbf{X}, t) = \frac{\partial}{\partial X_j} v_i\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big).$$

Here i, j are free indices.



Proof (Cont.)

The chain rule gives

$$\frac{\partial F_{ij}}{\partial t}(\mathbf{X},t) = \frac{\partial v_i}{\partial x_k}\big(\boldsymbol{\phi}(\mathbf{X},t),t\big)\frac{\partial \phi_k}{\partial X_j}(\mathbf{X},t)$$
$$= \frac{\partial v_i}{\partial x_k}\big(\boldsymbol{\phi}(\mathbf{X},t),t\big)F_{kj}(\mathbf{X},t).$$

Here i,j are free and k is a dummy index. In tensor form,

$$\frac{\partial F}{\partial t}(\mathbf{X},t) = \left[\nabla^{\mathrm{X}}\mathbf{v}\right]_m(\mathbf{X},t)F(\mathbf{X},t).$$

Hence

$$\left[\frac{\partial F}{\partial t}F^{-1}\right](\mathbf{X},t) = \left[\nabla^{\mathrm{X}}\mathbf{v}\right]_m(\mathbf{X},t).$$

Proof (Cont.)

Taking the trace gives

$$\text{tr}\left[\frac{\partial F}{\partial t}(\mathbf{X},t)F(\mathbf{X},t)^{-1}\right] = \left[\nabla^{\text{x}}\cdot\mathbf{v}\right]_m(\mathbf{X},t).$$

Substitution into equation (4.22) yields

$$\frac{\partial}{\partial t}\det\big(F(\mathbf{X},t)\big) = \det\big(F(\mathbf{X},t)\big)\left[\nabla^{\text{x}}\cdot\mathbf{v}\right]_m(\mathbf{X},t).$$

Equivalently,

$$\frac{\partial J}{\partial t} = J\left[\nabla^{\text{x}}\cdot\mathbf{v}\right]_m.$$



Reynolds Transport Theorem

Result 4.10 (Reynolds Transport Theorem)

Let

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

be a motion with associated spatial velocity $\mathbf{v}(\mathbf{x}, t)$. Let $\Omega_t \subseteq B_t$ be arbitrary, and suppose that its boundary $\partial\Omega_t$ is smooth with outward-pointing unit normal $\mathbf{n}$.

If q is an arbitrary spatial scalar field, then

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} q(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t} \{\dot{q}(\mathbf{x}, t) + q(\mathbf{x}, t) \nabla^{\mathrm{x}} \cdot \mathbf{v}(\mathbf{x}, t)\} \, \mathrm{d}V_{\mathrm{x}}.$$



Result 4.10 (Cont.)

Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} q(\mathbf{x},t)\,\mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t}\frac{\partial q}{\partial t}(\mathbf{x},t)\,\mathrm{d}V_{\mathrm{x}} + \int_{\partial\Omega_t} q(\mathbf{x},t)\mathbf{v}(\mathbf{x},t)\cdot\mathbf{n}(\mathbf{x},t)\,\mathrm{d}A_{\mathrm{x}}. \tag{4.23}$$

The first form uses the total time derivative $\dot{q}$. The second separates the local rate of change from transport across the moving boundary.

Proof.

By Result 4.8,

$$\int_{\Omega_t} q(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega} q_m(\mathbf{X}, t) J(\mathbf{X}, t)\, \mathrm{d}V_{\mathrm{X}}.$$

Therefore,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} q(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}} &= \frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega} q_m(\mathbf{X}, t) J(\mathbf{X}, t)\, \mathrm{d}V_{\mathrm{X}} \\
&= \int_{\Omega} \frac{\partial}{\partial t} \left[q_m(\mathbf{X}, t) J(\mathbf{X}, t) \right], \mathrm{d}V_{\mathrm{X}} \\
&= \int_{\Omega} \left\{ \frac{\partial q_m}{\partial t} J + q_m \frac{\partial J}{\partial t} \right\} \mathrm{d}V_{\mathrm{X}}.
\end{aligned}$$

Proof (Cont.)

Using

$$\left[\frac{\partial q_m}{\partial t}\right]_s (\mathbf{x}, t) = \dot{q}(\mathbf{x}, t)$$

and Result 4.8, the first term becomes

$$\int_\Omega \frac{\partial q_m}{\partial t}(\mathbf{X}, t) J(\mathbf{X}, t)\, \mathrm{d}V_{\mathrm{X}} = \int_{\Omega_t} \dot{q}(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}}.$$

Result 4.9 gives

$$\frac{\partial J}{\partial t}(\mathbf{X}, t) = J(\mathbf{X}, t)\left[\nabla^{\mathrm{x}} \cdot \mathbf{v}\right]_m (\mathbf{X}, t).$$

Proof (Cont.)

Using Result 4.9 and then Result 4.8 again,

$$\begin{aligned}
&\int_\Omega q_m(\mathbf{X},t)\frac{\partial J}{\partial t}(\mathbf{X},t)\,\mathrm{d}V_\mathrm{X} \\
&\quad = \int_\Omega q_m(\mathbf{X},t)\left[\nabla^\mathrm{X}\cdot\mathbf{v}\right]_m(\mathbf{X},t)J(\mathbf{X},t)\,\mathrm{d}V_\mathrm{X} \\
&\quad = \int_{\Omega_t} q(\mathbf{x},t)\nabla^\mathrm{x}\cdot\mathbf{v}(\mathbf{x},t)\,\mathrm{d}V_\mathrm{x}.
\end{aligned}$$

Consequently,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} q\,\mathrm{d}V_\mathrm{x} = \int_{\Omega_t}\left\{\dot{q}+q\nabla^\mathrm{x}\cdot\mathbf{v}\right\}\mathrm{d}V_\mathrm{x}.$$

This proves the first form of the theorem.

Proof (Cont.)

By Result 4.7, the total time derivative is

$$\dot{q}(\mathbf{x}, t) = \frac{\partial q}{\partial t}(\mathbf{x}, t) + \nabla^{\mathrm{x}} q(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t).$$

Thus

$$\begin{aligned} \dot{q} + q\nabla^{\mathrm{x}} \cdot \mathbf{v} &= \frac{\partial q}{\partial t} + \nabla^{\mathrm{x}} q \cdot \mathbf{v} + q\nabla^{\mathrm{x}} \cdot \mathbf{v} \\ &= \frac{\partial q}{\partial t} + \nabla^{\mathrm{x}} \cdot (q\mathbf{v}). \end{aligned}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} q \, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t} \frac{\partial q}{\partial t} \, \mathrm{d}V_{\mathrm{x}} + \int_{\Omega_t} \nabla^{\mathrm{x}} \cdot (q\mathbf{v}) \, \mathrm{d}V_{\mathrm{x}}.$$

Proof (Cont.)

The divergence theorem gives

$$\int_{\Omega_t} \nabla^{\mathrm{x}} \cdot (q\mathbf{v}) \, \mathrm{d}V_{\mathrm{x}} = \int_{\partial\Omega_t} q\mathbf{v} \cdot \mathbf{n} \, \mathrm{d}A_{\mathrm{x}}.$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} q(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t} \frac{\partial q}{\partial t}(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} + \int_{\partial\Omega_t} q(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x}, t) \, \mathrm{d}A_{\mathrm{x}},$$

which is equation (4.23). □



Boundary Integrals

Result 4.11 (Change of Variables on a Boundary)

Let $q(\mathbf{x}, t)$ be an arbitrary spatial scalar field, $\mathbf{w}(\mathbf{x}, t)$ an arbitrary spatial vector field, and $T(\mathbf{x}, t)$ an arbitrary spatial second-order tensor field.

Consider an arbitrary set $\Omega \subseteq B$ that maps to

$$\Omega_t = \boldsymbol{\phi}_t(\Omega) \subseteq B_t.$$

Let $\mathbf{N}(\mathbf{X})$ be the outward-pointing unit normal on $\partial\Omega$, and let $\mathbf{n}(\mathbf{x})$ be the outward-pointing unit normal on $\partial\Omega_t$.

Define the cofactor field

$$G(\mathbf{X}, t) := J(\mathbf{X}, t)F(\mathbf{X}, t)^{-T}.$$

Result 4.11 (Cont.)

The scalar boundary integral transforms according to

$$\int_{\partial\Omega_t} q(\mathbf{x},t)\mathbf{n}(\mathbf{x})\ \mathrm{d}A_\mathrm{x} = \int_{\partial\Omega} q_m(\mathbf{X},t)G(\mathbf{X},t)\mathbf{N}(\mathbf{X})\ \mathrm{d}A_\mathrm{X}.$$

The vector flux transforms according to

$$\int_{\partial\Omega_t} \mathbf{w}(\mathbf{x},t)\cdot\mathbf{n}(\mathbf{x})\ \mathrm{d}A_\mathrm{x} = \int_{\partial\Omega} \mathbf{w}_m(\mathbf{X},t)\cdot\big(G(\mathbf{X},t)\mathbf{N}(\mathbf{X})\big)\ \mathrm{d}A_\mathrm{X}.$$



Result 4.11 (Cont.)

The tensor boundary integral transforms according to

$$\int_{\partial\Omega_t} T(\mathbf{x},t)\mathbf{n}(\mathbf{x})\,\mathrm{d}A_x = \int_{\partial\Omega} T_m(\mathbf{X},t)G(\mathbf{X},t)\mathbf{N}(\mathbf{X})\,\mathrm{d}A_X.$$

Proof.

See any good book on advanced calculus. □



Volume-Preserving Motions



Definition (Volume-Preserving Motion)

A motion

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

is called *volume-preserving*, or *isochoric*, if

$$\text{Vol}(\Omega_t) = \text{Vol}(\Omega)$$

for every $t \geq 0$ and every $\Omega \subseteq B$, where, as usual,

$$\Omega_t = \boldsymbol{\phi}_t(\Omega).$$

Result 4.12 (Characterization of Volume Preservation)

Let

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

be a motion with spatial velocity $\mathbf{v}$ and deformation gradient

$$F(\mathbf{X}, t) = \nabla^{\mathrm{X}} \boldsymbol{\phi}(\mathbf{X}, t).$$

Then $\boldsymbol{\phi}$ is volume-preserving if and only if

$$J(\mathbf{X}, t) = \det\big(F(\mathbf{X}, t)\big) = 1, \qquad \mathbf{X} \in B, \quad t \geq 0,$$

or, equivalently, if and only if

$$\nabla^{\mathrm{x}} \cdot \mathbf{v}(\mathbf{x}, t) = 0, \qquad \mathbf{x} \in B_t, \quad t \geq 0.$$



Proof.

The result follows from the observations

$$\begin{aligned} \text{Vol}(\Omega_t) &= \int_{\Omega_t} \mathrm{d}V_{\mathrm{x}} \\ &= \int_{\Omega} J(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}} \end{aligned}$$

and, by the Reynolds transport theorem,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\text{Vol}(\Omega_t) = \int_{\Omega_t} \nabla^{\mathrm{x}} \cdot \mathbf{v}(\mathbf{x}, t)\ \mathrm{d}V_{\mathrm{x}}.$$

Applying the localization theorem to every $\Omega \subseteq B$ and every $\Omega_t \subseteq B_t$ gives the two pointwise characterizations. The details are left as an exercise. □



The Leibniz Integral Rule

Example (The Leibniz Integral Rule)

Let $a, b \in C^1([0, T])$ satisfy

$$a(t) < b(t), \qquad 0 \le t \le T,$$

and suppose that $f(x, t)$ and $\partial f / \partial t$ are continuous on the moving interval

$$\Omega_t = (a(t), b(t)).$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{a(t)}^{b(t)} f(x, t) \, \mathrm{d}x = \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) \, \mathrm{d}x + f(b(t), t) b'(t) - f(a(t), t) a'(t).$$

We derive this formula from the Reynolds transport theorem.

Example (The Leibniz Integral Rule, Cont.)

Take the fixed reference interval $\Omega = (0,1)$ and define the motion

$$\phi(X,t) = a(t) + \big(b(t) - a(t)\big)X, \qquad 0 < X < 1.$$

Then

$$\phi(0,t) = a(t), \qquad \phi(1,t) = b(t), \qquad \phi(\Omega,t) = \Omega_t.$$

The spatial velocity satisfies

$$\begin{aligned} v(\phi(X,t),t) &= \frac{\partial \phi}{\partial t}(X,t) \\ &= a'(t) + \big(b'(t) - a'(t)\big)X. \end{aligned}$$

Therefore, the two endpoint velocities are

$$v(a(t),t) = a'(t), \qquad v(b(t),t) = b'(t).$$

Example (The Leibniz Integral Rule, Cont.)

In one space dimension, the outward unit normals are

$$n(a(t)) = -1, \qquad n(b(t)) = +1.$$

The boundary integral is the sum over the two endpoints. Hence

$$\begin{aligned}\int_{\partial\Omega_t} fvn \, \mathrm{d}A &= f(b(t),t)v(b(t),t)(+1) \\ &\quad + f(a(t),t)v(a(t),t)(-1) \\ &= f(b(t),t)b'(t) - f(a(t),t)a'(t).\end{aligned}$$

The signs come from the two outward normals.

Example (The Leibniz Integral Rule, Cont.)

The boundary form of the Reynolds transport theorem gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} f(x,t)\,\mathrm{d}x &= \int_{\Omega_t}\frac{\partial f}{\partial t}(x,t)\,\mathrm{d}x + \int_{\partial\Omega_t} fvn\,\mathrm{d}A \\ &= \int_{a(t)}^{b(t)}\frac{\partial f}{\partial t}(x,t)\,\mathrm{d}x \\ &\quad + f(b(t),t)b'(t) - f(a(t),t)a'(t).\end{aligned}$$

Thus Reynolds transport in one dimension is precisely the Leibniz integral rule for moving endpoints.

If an endpoint is fixed, its velocity is zero and its boundary term vanishes.